

KY 1013 BRIDGE
Spanning Sand Lick Creek ^{at} ~~under~~ Route KY 1013
Plummers Landing
Fleming County
Kentucky

HAER No. KY-29

HAER
KY-29

PHOTOGRAPHS

WRITTEN HISTORICAL AND DESCRIPTIVE DATA

Historic American Engineering Record
National Park Service
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Department of the Interior
Atlanta, Georgia 30303

HISTORIC AMERICAN ENGINEERING RECORD

KY 1013 BRIDGE

HAER No. KY-29

Location: Spanning Sand Lick Creek near Plummers Landing
Fleming County, Kentucky

UTM Northing: 16 4243940 Easting: 16 274580

Date of Construction: 1893

Present Owner: Kentucky Transportation Cabinet

Significance: The Pratt Half Hipp Pony Truss was constructed in 1893 by the
Pittsburg Bridge Company. It is the only documented bridge in the
state known to have been constructed by this company.

Historian Jayne Henderson Fiegel

KY 1013 BRIDGE
HAER No. KY-29 (page 2)

The KY 1013 bridge over Sand Lick has been determined eligible for the National Register of Historic Places.

The KY 1013 bridge crosses Sand Lick Creek in the eastern part of Fleming County near the tiny community of Plummers Landing. The structure is situated about five miles east of the Rowan County line and approximately fourteen miles southeast of Flemingsburg, the county seat. Sand Lick Creek is drained by the Licking River, a major waterway in northeastern Kentucky. KY 1013 is a two-lane, secondary, state route.

The Sand Lick Creek bridge is a Pratt Half Hipp Pony Truss constructed in 1893 by the Pittsburg Bridge Company. It is the only documented bridge in the state by this particular bridge company.

The "Survey of Truss, Suspension and Arch Bridges in Kentucky" completed in January 1982 located 122 Pratt Half Hipp Ponies in the state. Ninety-eight percent of these structures, including the KY 1013 bridge, are pin-connected. Most of the Pratt Half Hipp Ponies were constructed before 1920 and the average length of the trusses was 58 feet. Thirty of these trusses have identifying builder/date plates representing the work of seven separate bridge builders or companies. The Champion Bridge Company of Wilmington, Ohio built twenty-one of these structures and six other companies built the other nine bridges with extant builder plates.

Significance of National Register Eligible Property

The KY 1013 bridge is eligible for the National Register of Historic Places as a structure of state and local importance which possesses integrity of location, design, setting, materials and workmanship and embodies the distinctive characteristics of a type, period and method of construction. The bridge, although in poor structural condition, is a good example of the evolution of civil engineering technology and bridge construction in Kentucky.

Although Kentucky has bridges representing most of the early truss designs, beginning with the wooden king post and queen post and Town lattice trusses and including the mid-19th century metal truss designs of the Bowstring Arch and Whipple trusses, the first widespread bridge construction efforts in Kentucky did not materialize until the late 19th century. During this period the Pratt and Warren trusses accounted for a great majority of bridges constructed. As bridge designs evolved, many later truss improvements retained the Pratt configuration of compression and tension member while changing the shape of the top chords. The Parker truss, for example, is clearly a Pratt with a polygonal top chord. Because of the arched top chord, the Parker is stronger than a regular Pratt, but uses nearly the same amount of materials. A further modification on the Parker is the camelback, which has a polygonal top chord of exactly five slopes.

In the latter part of the 19th century, increasing railroad weight requirements necessitated new truss developments capable of withstanding greater loads. A major advance in strengthening the metal truss came with implementation of sub-struts and sub-ties to

stiffen the truss under moving loads. The Baltimore and Pennsylvania trusses, named after railroad lines that used them extensively, utilized these components with either a straight or arched top chord, respectively. Early examples of concrete arch, suspension and cantilever bridges are also found in Kentucky.

The most popular late 19th and early 20th century bridge type was the Pratt truss. This truss type was patented in 1844 by Thomas and Caleb Pratt and utilized wooden compression posts and wrought iron tension members. Soon all members were constructed of metal, first cast and wrought iron and then steel. The Pratt Through Truss was capable of spanning longer distances and bearing greater loads.

Pratt Pony Trusses, low pony trusses without top lateral bracing, were used for economical reasons. When the requirements for weight and length of a particular crossing could be met by support spans of limited weight capacity, then Pratt Ponies were utilized.

On all truss spans the end posts and top chord act in compression with the bottom chord in tension. In a truss, unlike a rigid arch, at least one bearing point at the abutment must be able to expand or move. In the Pratt truss, the verticals between the end posts go into compression to keep the top chord from collapsing, and the diagonals act in tension to support the deck. However, the first vertical member (hip-vertical) next to the inclined end post must be placed in tension to support the deck when a load first enters the bridge.

Pratt trusses have both diagonals and counters in the web system acting in tension. Inclined members that are not parallel to the nearest end post are called diagonals. Inclined members parallel to the nearest end post are called counters. Diagonals support the dead load of bridge weight and the live load of traffic. Counters support only the live load of the bridge. Counters always intersect with a diagonal between two panel points (or floor beams) of the bridge.

Counters accept or counteract the live load support from the diagonals as a load moves across the bridge. Tension support for the deck goes from a diagonal past the compression post, which is released as a load passes, to the next counter or diagonal. When a load passes an interior compression post and it is released, the compressive stress is thrown into the adjacent compression posts or end posts to keep the top and bottom chords apart.

The compression members of a Pratt truss must be rigid and of sturdy construction. On the KY 1013 bridge, the end post and top chords are constructed of 2 channels, cover plate and stay bars. The intermediate posts are paired angles and lattice bars. The main tension members on the KY 1013 bridge are as follows: bottom chords are 2 rectilinear and two round eye bars, loop welded, diagonals are two rectilinear and two loop welded round eyebars.

The floor system of the KY 1013 bridge has rolled I-beam floor beams and wood stringers. The structure has a wood deck, and the truss is supported by rough cut stone abutments.

The KY 1013 bridge over Sand Lick Creek is eligible for the National Register of Historic Places under Criteria C. The

structure is an early representative of an engineering type of technology popular throughout Kentucky during the last quarter of the 19th and first part of the 20th centuries. Constructed in 1893, it is the only structure built by the Pittsburg Bridge Company in the state.